

SmartCEM Central Data Controller

Installation Manual

CODEL

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CODEL International Ltd. is a UK company based in Bakewell, Derbyshire, specialising in the design and manufacture of high technology instrumentation for monitoring combustion processes and atmospheric pollution emissions. The company philosophy is simply, to design well-engineered, rugged and reliable equipment, capable of continuous operation over long periods with minimal maintenance.

While every effort has been made to ensure that this manual is free from errors, CODEL supply all information without warranty.

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Table of Contents

1. CODEL Central Data Controller (CDC) – Introduction	1
1.1. Foreword	1
1.2. Central Data Controller	1
1.3. User Interface	1
1.4. Analogue and Logic Outputs	2
1.5. Inputs	2
1.6. CDC Protection	3
2. Summary Specification	4
3. Installation	5
4. Wiring Connections	6

1. CODEL Central Data Controller (CDC) – Introduction

Please take a few minutes to read this section before proceeding with installation. It is designed to provide an overview of the SmartCEM system and its capabilities so that the information given later will be more easily understood.

1.1. Foreword

SmartCEM is the ultimate solution for continuous monitoring of stack emissions. It is a fully integrated monitoring concept from the basic analysers and their calibration verification packages through to digital communications, data logging and automatic reporting.

At the heart of this concept is the SmartCEM station containing all the analysers and monitors to provide comprehensive stack emission monitoring. The SmartCEM Station Control Unit (SCU) provides power to and communicates with the analysers and monitors within that station.

Data from up to 32 SmartCEM stations is transmitted via a serial digital link (CODEL SmartBUS) to the Central Datapoint where it is logged on a dedicated pc or assembled for onward transmission to a plant computer or DCS.

1.2. Central Data Controller

The CDC derives its power from the Power Supply Unit. Power input to the CDC is 48V DC from the PSU.

1.3. User Interface

The analyser is supplied with SmartCOM pc software that enables a pc connected to the CDC to display all real-time measurements as well as the setting-up and display of all the analysers' configurable operating parameters. This interface includes a diagnostic mode that allows interrogation of the condition of the analysers' critical functions for service and trouble-shooting purposes.

1.4. Analogue and Logic Outputs

The CDC can be equipped with an analogue output (0/4-20mA) for measurement channels as required. Each output is freely configurable for any span (within the limits for that measurement) and measurement unit (vpm or %, mg/Nm³ or mass emissions if a flow monitor is present in the system).

Volt-free contact outputs (110V DC/125V AC @ 1A max) can be provided for data valid and measurement alarm level (common output for all measurement channel alarms).

1.5. Inputs and Outputs (factory default settings)

Output Configuration

Current Outputs

Source	Address (Scu)	Type	Range	Response
1 Opacity	#1	4-20ma	0 -> 100.00%	Analyser
2 Extinction	#1	4-20ma	0 -> 0.2000	Analyser
3 Flow	#1	4-20ma	0 -> 10.0m/S	Analyser
4 Flow	#1	4-20ma	0 -> 10.0m/S	Analyser

Relay Outputs

Source	Address	Function	Response
1 Scu	#1	Scu Comms (Power Fail)	30seconds
2 Scu	#1	Dcem Data Valid	Immediate
3 Scu	#1	Maintainance	Immediate
4 Scu	#1	Dcem Calibration	Immediate
5 Scu	#1	Scu Comms (Power Fail)	30seconds
6 Scu	#1	Vcem Data Valid	Immediate
7 Scu	#1	Maintainance	Immediate
8 Scu	#1	Vcem Calibration	Immediate

Cdc Input Configuration

Contact Inputs

Source	Address	Function
1 Gcem Scu #1		Span Check Co]
2 Gcem Scu #1		Span Check No] Reserved For Future Use
3 Gcem Scu #1		Span Check So2]
4 Dcem Scu #1		Span Check
5 Vcem Scu #1		Span Check
6 Not Implemented		
7 Not Implemented		
8 Not Implemented		

1.6. CDC Protection

The unit is constructed to IP67 standards and is designed for ambient temperatures from -20° to +55°C.

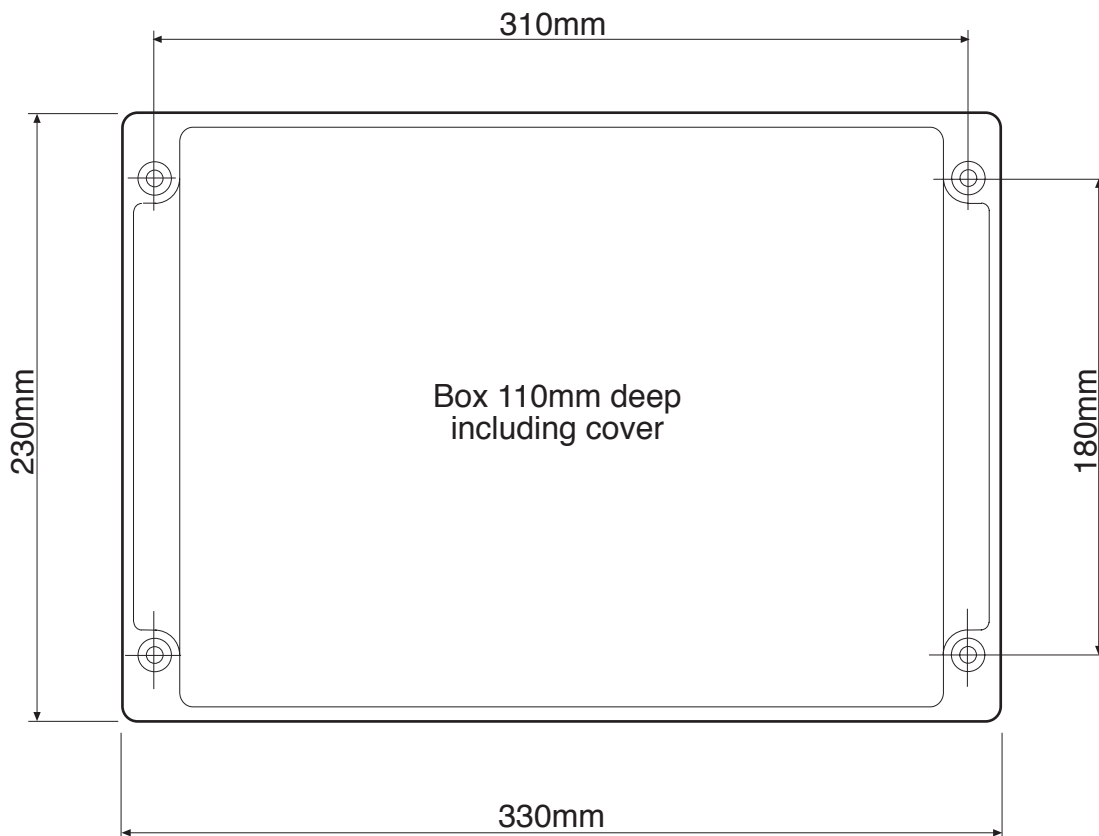
2. Summary Specification

Construction	Die-cast aluminium protected to IP67
Ambient Temperature	-20° to +55°C
Power Input	48V DC from the PSU

3. Installation

To mount the CDC first remove the cover by loosening the four captive screws. Secure the case to a firm support by use of the four mounting holes, one in each corner. Since the mounting holes are located outside the seal of the case it is not necessary to seal the mounting holes after installation, nor is it necessary to remove the circuitry from the case for installation.

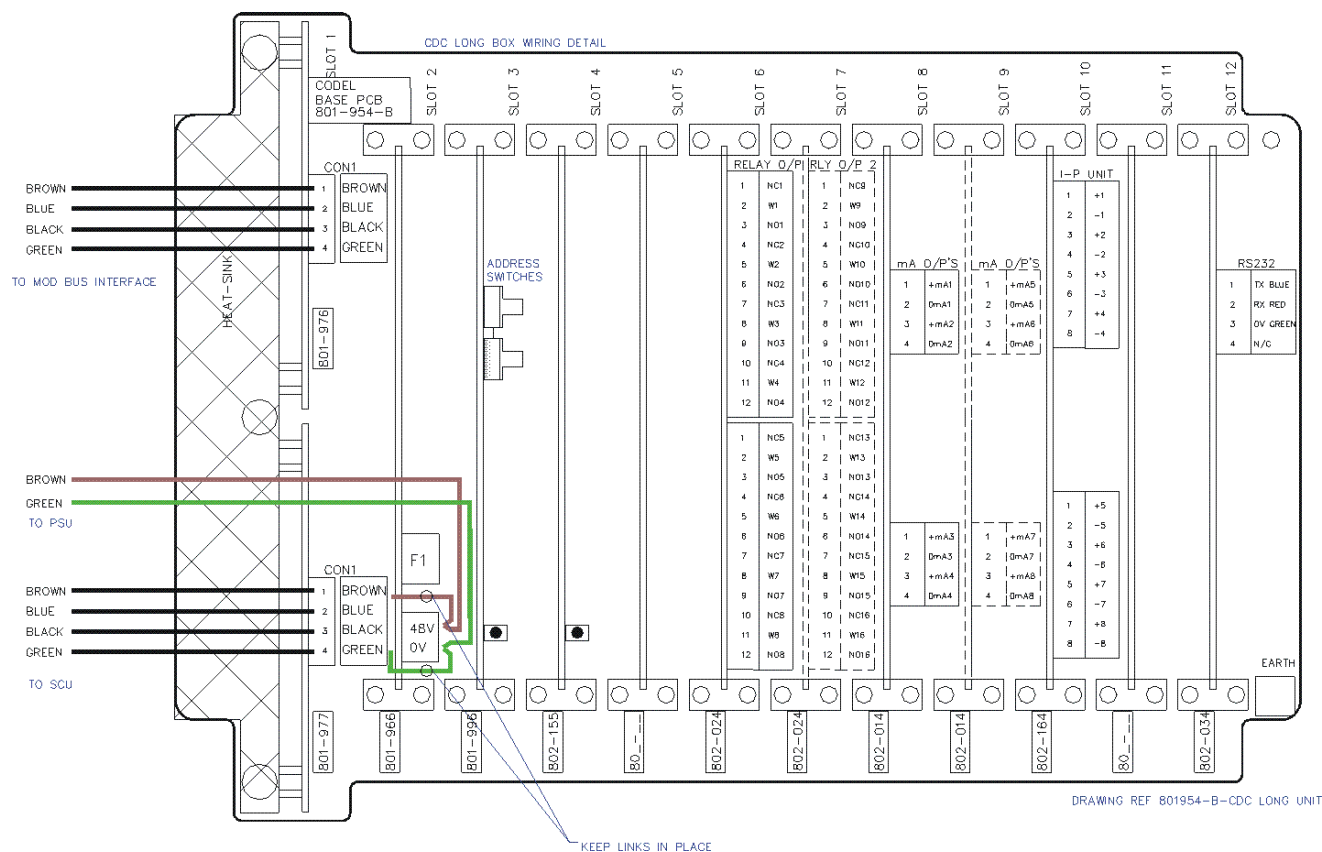
If commissioning is not to be carried out immediately, reattach the lid to the processor. Dimensions and mounting details are shown below.



CDC Dimensions and Mounting Hole Details

4. Wiring Connections

The wiring connections for the Central Data Controller are shown in the following illustrations.



CDC Wiring Connections

